

LEGACY MIGRATION RISK ASSISTANT

IBM BOB 2.0 HACKATHON | lablab.ai | BUG SQUASH

LEGACY SQL SERVER

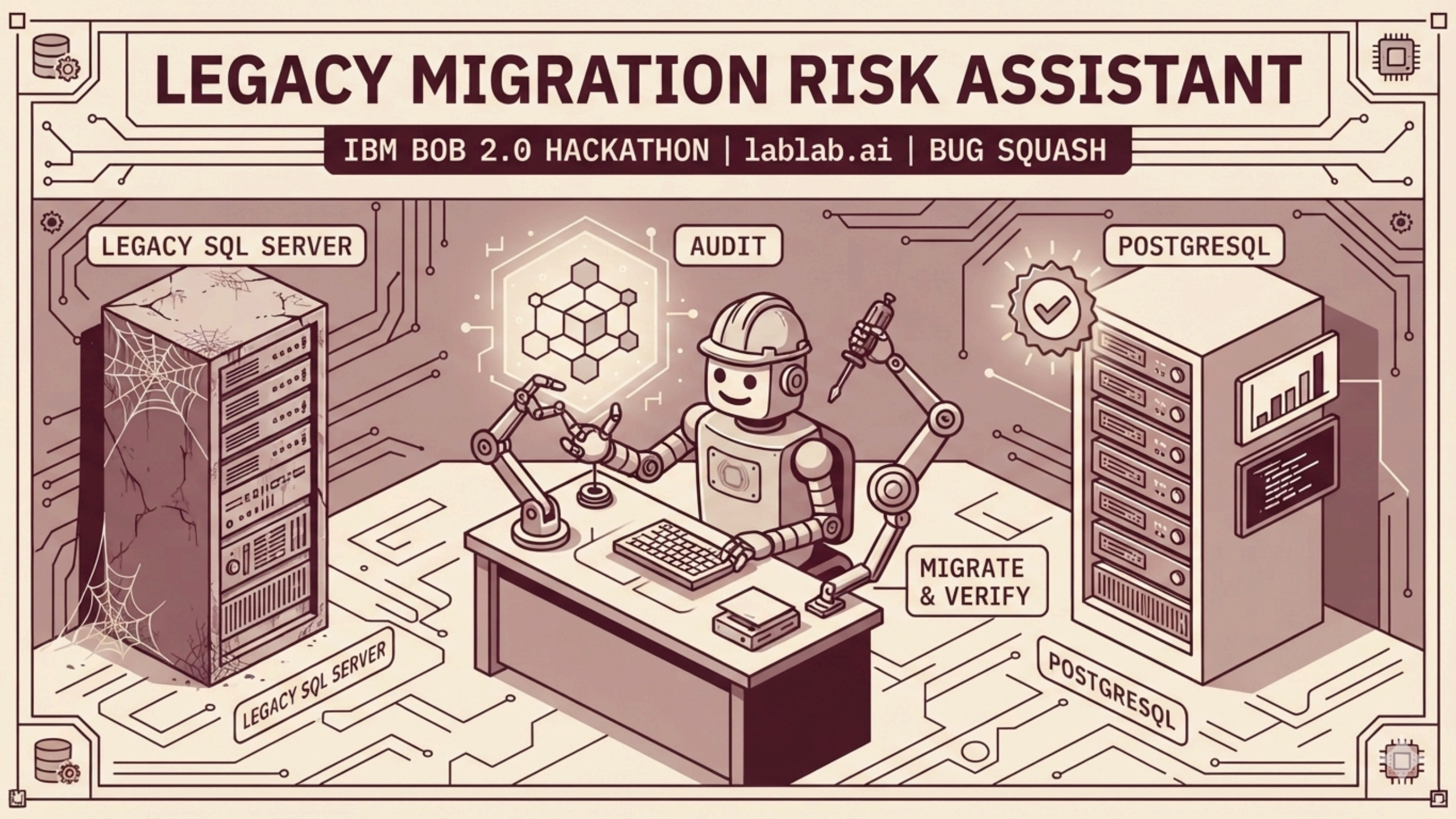
AUDIT

POSTGRESQL

MIGRATE
& VERIFY

LEGACY SQL SERVER

POSTGRESQL



Every Team Inherits a Database Nobody Wants to Touch

Legacy modernization is high-risk, low-glamour, and usually pushed to "next quarter"

- ▶ A real, widespread enterprise challenge — not a synthetic demo. We used Microsoft's own official 1998-era Northwind & Pubs schemas, genuinely deprecated, still common in production estates today
- ▶ Deprecated syntax, undocumented business logic buried in stored procedures and triggers
- ▶ Manually auditing migration risk typically takes a senior engineer **a full working day, per database**
- ▶ Most modernization tools stop at a risk report — a to-do list, and nothing actually moved
- ▶ We wanted to close that gap: audit, score, **and actually migrate**

Legacy SQL Server → Bob → Postgres

One real, working pipeline — not a diagram of one

Legacy SQL Server

- Northwind (1998)
- Pubs (1998)
- 13 procs/triggers
- 16 views
- No docs, no tests



IBM Bob 2.0

- Deprecated-syntax scan
- Business-logic extraction
- Risk scoring w/ evidence
- DDL + data-copy generation



PostgreSQL

- 13 tables + 13 FKs created
- 3,308 rows migrated
- Row counts verified
- Exact match to source



Fully containerized

SQL Server 2019 + PostgreSQL 16, both Docker — reproducible, zero local install



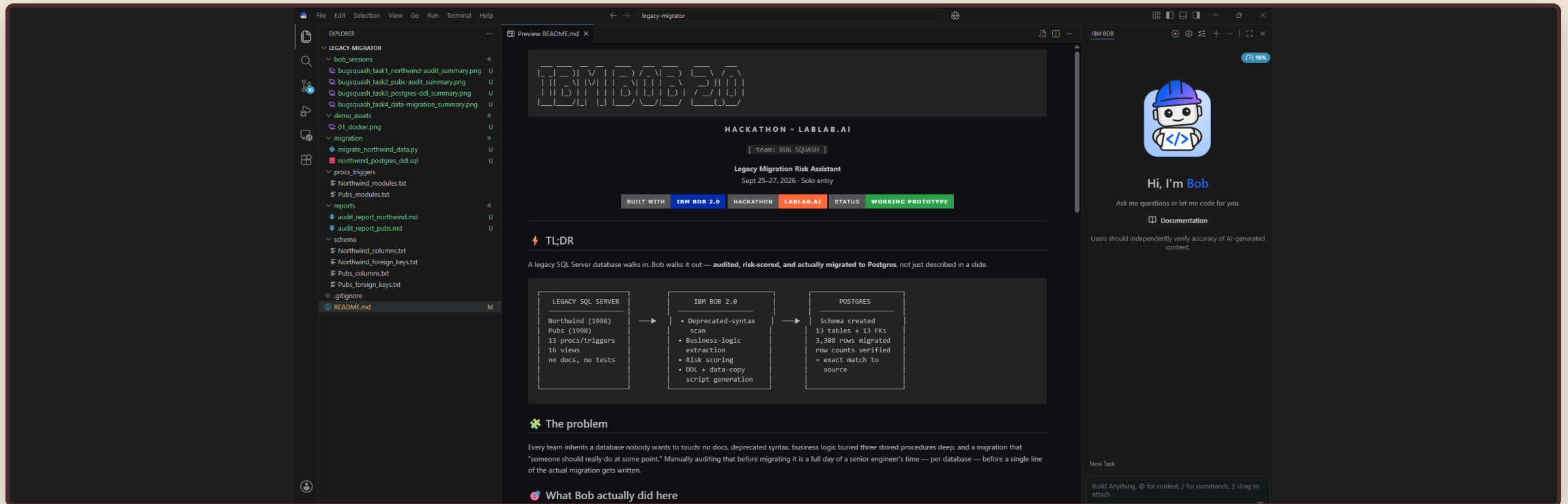
~160x productivity gain

8-hour manual audit → under 3 minutes, automated

Four Distinct Jobs, One Real Pipeline

Audit · Generate DDL · Migrate · Verify

Bob IDE for interactive review · Bob Shell CLI (API key) for scripted runs · Microsoft's public sample data only — no proprietary information



Bob IDE — project loaded, ready to work

Mapping the Legacy Schema

Bob's dependency-graph analysis of Northwind's foreign keys

The screenshot displays the Legacy Migration Risk Assistant interface, which is used for analyzing legacy database schemas and generating migration reports. The interface is divided into several sections:

- Left Panel:** Contains a file explorer and a terminal window. The terminal shows the command `show suggestions` being executed.
- Center Panel:** Displays a migration audit report for the Northwind Legacy SQL Server. The report includes an executive summary and a list of deprecated/legacy SQL Server syntax findings.
- Right Panel:** Shows a dependency graph analysis of Northwind's foreign keys. It includes a table of root tables (no inbound FKs) and a table of second-tier tables (one FK level deep).

Root tables (no inbound FKs)

Table	PK	Notable columns
Categories	CategoryID (identity)	CategoryName, Description, Picture (bytea)
Customers	CustomerID (char 5)	Company/contact info, address fields
CustomerDemographics	CustomerTypeID (char 10)	customerDesc (free text)
Employees	EmployeeID (identity)	HR fields, Photo (bytea), ReportsTo (self-FK)
Region	RegionID (identity)	RegionDescription
Shippers	ShipperID (identity)	CompanyName, Phone
Suppliers	SupplierID (identity)	Company/contact info, HomePage

Second-tier tables (one FK level deep)

Table	PK	FK references
Territories	TerritoryID (varchar 20)	→ Region.RegionID
Products	ProductID (identity)	→ Categories.CategoryID, → Suppliers.SupplierID

Third-tier tables (two FK levels deep)

Table	PK	FK references
Orders	OrderID (identity)	→ Customers.CustomerID, → Employees.EmployeeID, → Shippers.ShipperID

Junction / leaf tables

Table	PK	FK references
CustomerCustomerDemographics	(CustomerID, CustomerTypeID)	→ Customers.CustomerID, → CustomerDemographics.CustomerTypeID
EmployeeTerritories	(EmployeeID, TerritoryID)	→ Employees.EmployeeID, → Territories.TerritoryID
Order Details	(OrderID, ProductID)	→ Orders.OrderID, → Products.ProductID

Bob reviewing migration. Table formatting is cool 😎

Legacy Migration Risk Assistant

Bug Squash

Bob, Executing Live

Reading files, running tool calls, doing the actual work

The screenshot displays the Legacy Migration Risk Assistant (Bob) interface. The main window shows a migration audit report for the Northwind Legacy SQL Server to PostgreSQL Migration. The report includes an executive summary and a list of deprecated/legacy SQL Server syntax findings. A red dashed arrow points from the text 'Bob Summarizing tables with dependency graph' to the 'Key structural notes' section of the dependency graph.

Bob Summarizing tables with dependency graph

Key structural notes

- Employees has the only self-referencing FK (ReportsTo → EmployeeID), making it a circular dependency that a plain FK-ordered insert can't resolve without disabling constraints first.
- Order Details (note the space in the name) is the deepest leaf — it sits 4 levels below the roots (Categories/Suppliers → Products → Orders → Order Details).
- 7 of 13 tables use GENERATED ALWAYS AS IDENTITY PKs, requiring OVERRIDING SYSTEM VALUE on insert and a post-migration sequence resync.
- CustomerCustomerDemo and EmployeeTerritories are pure many-to-many junction tables with no non-key columns.

Dependency graph

Junction / leaf tables

Table	PK	FK references
CustomerCustomerDemo	(CustomerID, CustomerTypeID)	→ Customers.CustomerID, → CustomerDemographics.CustomerTypeID
EmployeeTerritories	(EmployeeID, TerritoryID)	→ Employees.EmployeeID, → Territories.TerritoryID
Order Details	(OrderID, ProductID)	→ Orders.OrderID, → Products.ProductID

Legacy Migration Risk Assistant

OUTLINE

Bob Findings

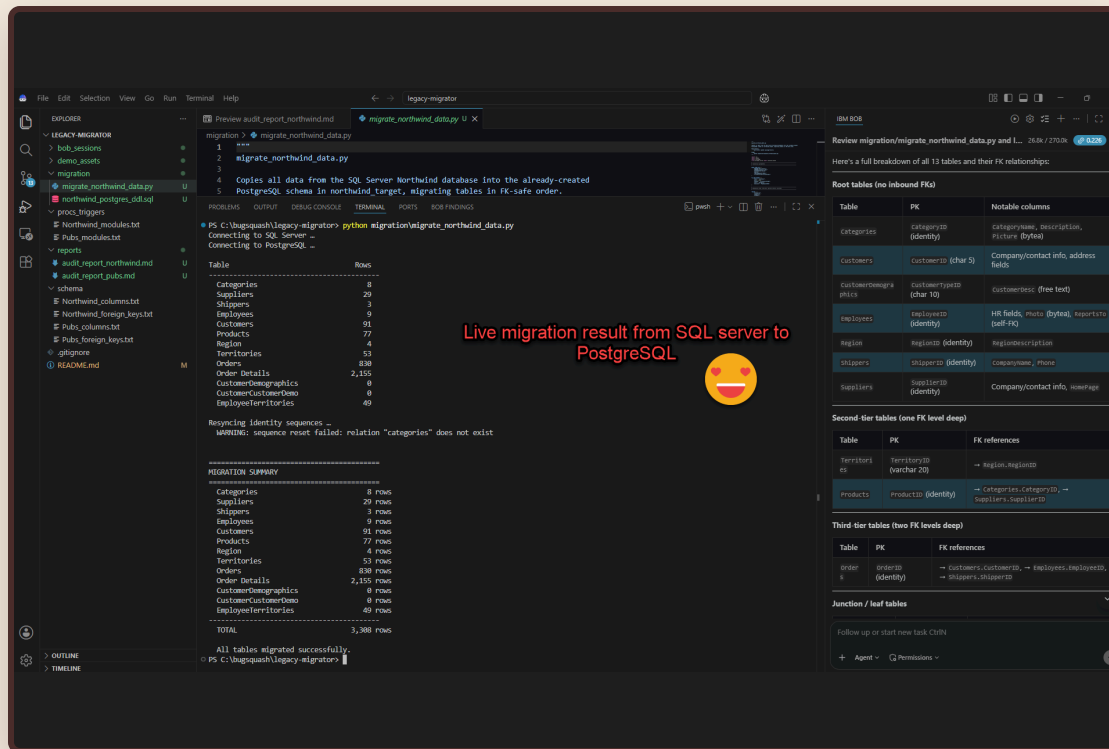
Ln 1, Col 1, Spaces 4 - UTF-8 - CRLF - Markdown - Bob Settings

Bug Squash

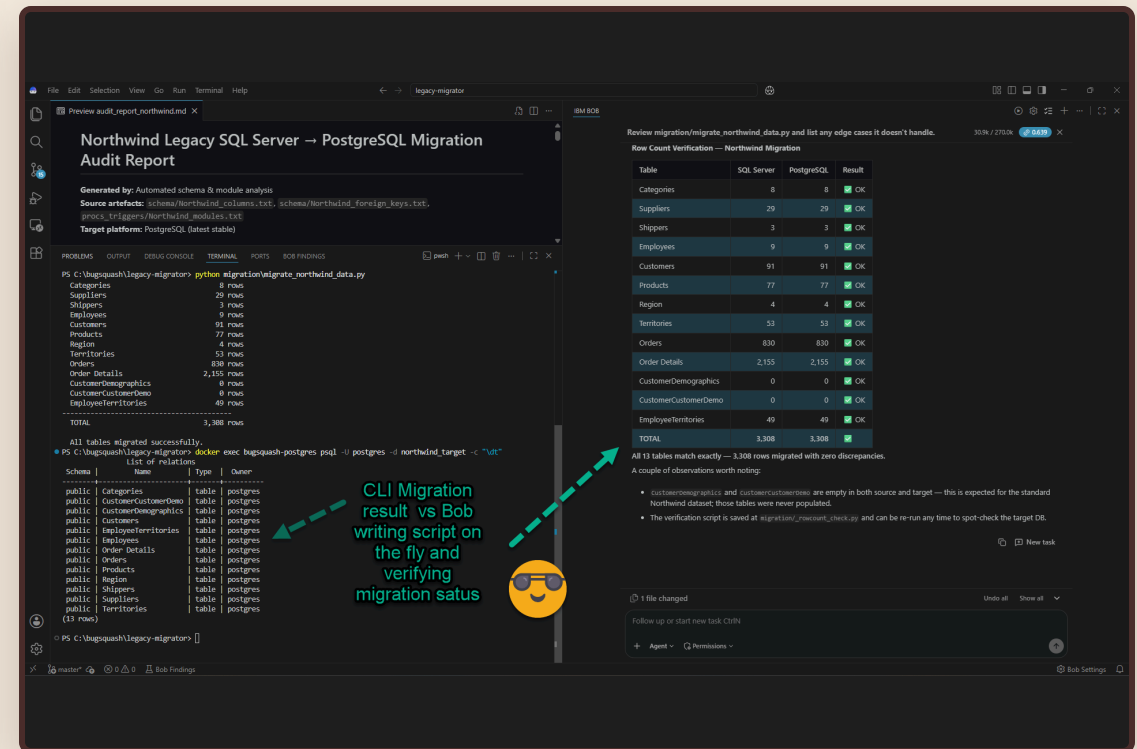
LIVE PROOF

Not Just Generated — Actually Executed

The migration ran, and Bob checked its own result



Migration script running — 3,308 rows moving into Postgres



Bob independently verifying the row counts match the source

Evidence-Based, Not Vague Warnings

Every risk score cites the exact offending line

Object	Type	Risk	Evidence
Ten Most Expensive Products	Proc	 5	SET ROWCOUNT 10 unsupported on Postgres
Invoices	View	 4	String concat + CONVERT (money , . . .)
Sales by Year	Proc	 3	DATENAME is T-SQL only
Employee Sales by Country	Proc	 2	Space-name only; otherwise ANSI
Alphabetical list of products	View	 1	SELECT *; space-name only

An 8-Hour Audit Became a 3-Minute Pass

And this time, the fix actually shipped

3,308

ROWS MIGRATED &
VERIFIED

13 / 13

TABLES & FOREIGN KEYS

~160x

FASTER THAN MANUAL
AUDIT

2x

VERIFIED — WIPED & RE-
RUN CLEAN

- Most legacy-audit tools stop at a report. This one closes the loop: Bob generated the migration, ran it, then independently verified its own result.

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github.com/viveknz/legacy-migrator

